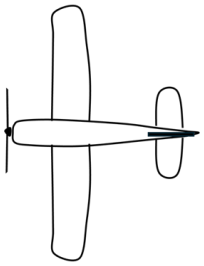
Problem-Based Learning

Wing aerodynamics and geometry

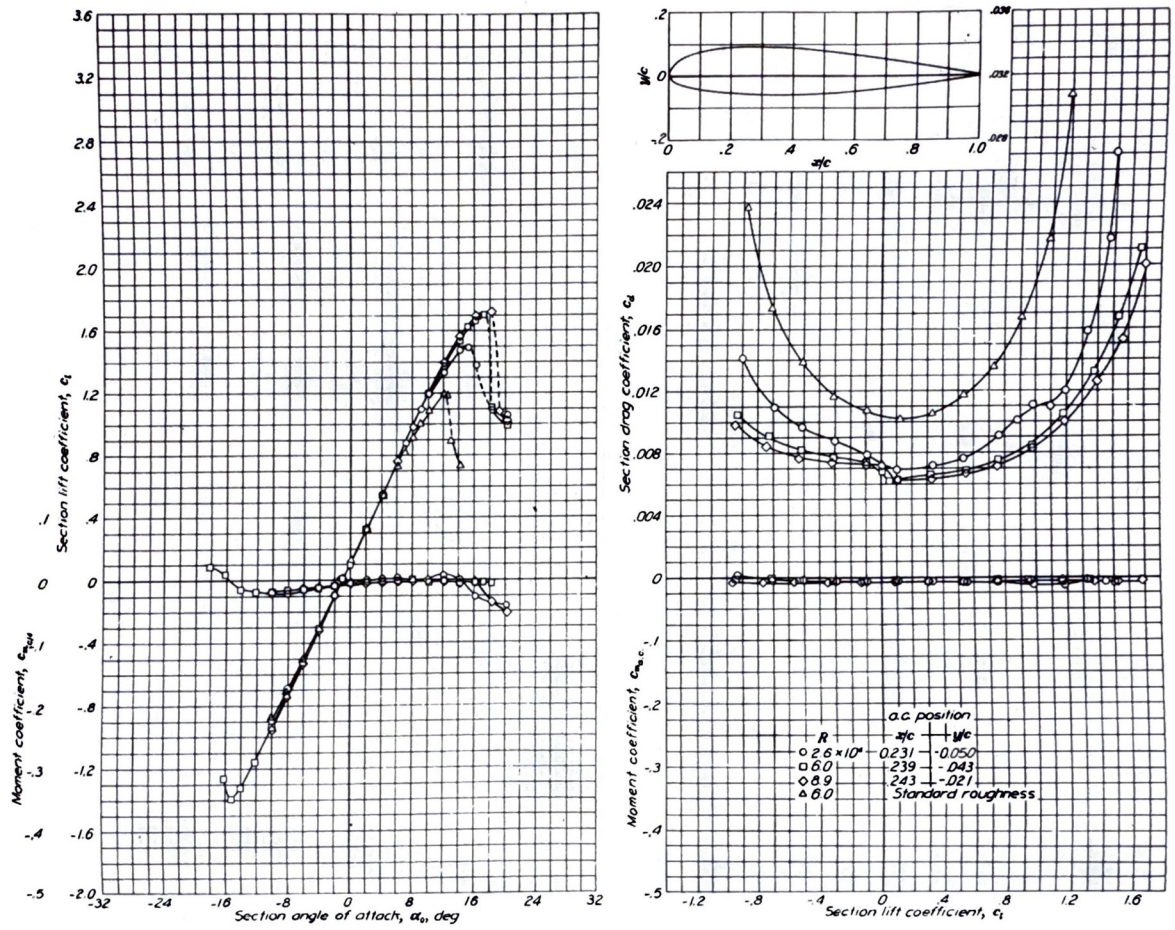
Question 1 **Past Exam: EMS611U 2025–26, Question 2(c)** **[12 marks]**



A tractor-configured, four-seat general aviation aircraft suitable for training purposes, powered by a piston engine with a fixed-pitch propeller. The aspect ratio of the aircraft's wing is $A = 8$.

(c) For the design under consideration, assume that the **NACA 23015** aerofoil is considered, with its data provided in Figure 1. At a Reynolds number of 2.6×10^6, determine: the stall angle of attack, the zero-lift angle of attack and the two-dimensional lift-curve slope. Calculate the 3D lift-curve slope and the stall speed for steady level flight, given that the aircraft mass is **950 kg** and the wing reference area is **16 m**^{2}.

You may take the air density as $1.225\,\mathrm{kg/m^3}$, the Oswald efficiency factor as 1 and gravity as $9.8\,\mathrm{N/kg}$.



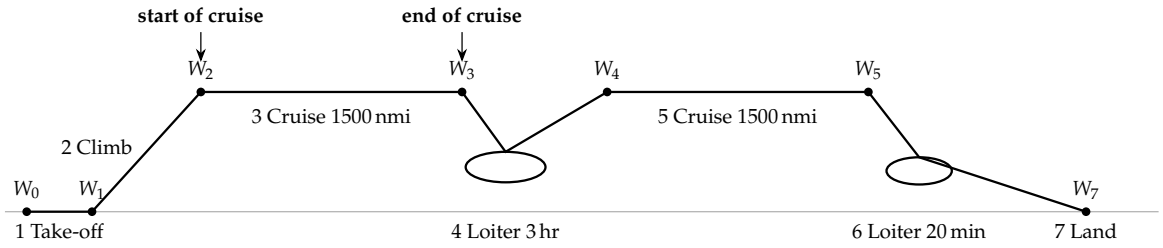
- $Re = 2.6\ 10^6$
- $Re = 6.0\ 10^6$
- $Re = 8.9\ 10^6$
- $Re = 6.0\ 10^6$, standard roughness

Figure 1. NACA 23015 aerofoil data.

Question 2 — Cruise lift coefficient

For the ASW design example, what is the lift coefficient at the **beginning** of cruise? At the **end**?

ASW mission and sizing results (from the take-off weight sizing example)



Sized take-off weight	$W_0 = 56{,}706\,\text{lb}$	Cruise	M 0.6 at 30,000 ft
Warm-up and take-off	$W_1/W_0 = 0.970$	Speed of sound / speed	$a = 994.8\,\text{ft/s},\ V = 596.9\,\text{ft/s}$
Climb	$W_2/W_1 = 0.985$	Density at 30,000 ft	$\rho = 8.907 \times 10^{-4}\,\text{slug/ft}^3$
Cruise 1500 nmi	$W_3/W_2 = 0.8581$	Take-off wing loading	$W_0/S = 88\,\text{lb/ft}^2$

The wing loading is taken from the existing aircraft that flies this mission, the Lockheed S-3 Viking (52,539 lb / 598 $\text{ft}^2 \approx 88\,\text{lb/ft}^2$).

Useful relations

$$L = W = q\,S\,C_L, \qquad q = \tfrac{1}{2}\rho V^2 \quad \Rightarrow \quad C_L = \frac{W/S}{q}$$

Question 3 — Wing planform selection

Select wing planform parameters (aspect ratio, taper, sweep, thickness, dihedral, etc.) for the ASW design example. Make reasonable guesses where needed, based on the characteristics of the current aircraft that performs the same mission, the **Lockheed S-3 Viking**. Use the wing area from Question 2 to find the span, root and tip chords and the mean aerodynamic chord.

Lockheed S-3A Viking (existing ASW aircraft)

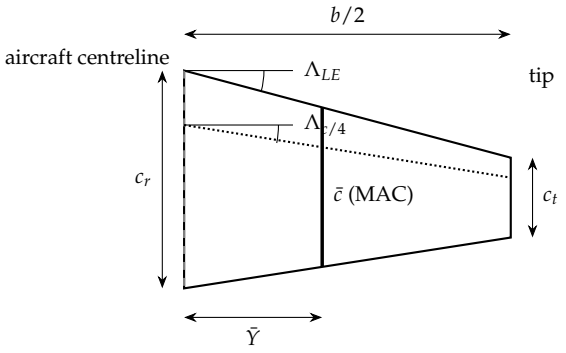
Wing span	68.7 ft (20.9 m)
Wing area	598 ft^2 (55.6 m^2)
Wing position	high (shoulder) wing
Leading-edge sweep	$\approx 15^\circ$, near-straight TE
Aerofoil	NACA 00-series mod.:
	16.3% root, 12% tip
Max take-off weight	52,539 lb
Max / cruise speed	429 kn / 350 kn
Engines	2 × TF34 turbofans

ASW requirements: cruise M 0.6 at 30,000 ft; long loiter (3 hr) at $(L/D)_{max}$; $W_0 \approx$ 56,700 lb; $L/D_{max} = 16$ assumed in sizing; take a maximum (dash) Mach number $M_{max} \approx 0.75$.

Planform definitions

$$A = \frac{b^2}{S}, \qquad \lambda = \frac{c_t}{c_r}, \qquad c_r = \frac{2S}{b\,(1+\lambda)}$$

$$\bar{c} = \frac{2}{3}\,c_r\,\frac{1+\lambda+\lambda^2}{1+\lambda}, \qquad \bar{Y} = \frac{b}{6}\,\frac{1+2\lambda}{1+\lambda}$$



Question 4 **Tails as "little wings"**

If tails are "little wings", why don't they have the same planform parameters as the airplane's wing?

Typical planform values

	Horizontal tail		Vertical tail	
	A	λ	A	λ
Fighter	3–4	0.2–0.4	0.6–1.4	0.2–0.4
Sailplane	6–10	0.3–0.5	1.5–2.0	0.4–0.6
Others (incl. transports)	3–5	0.3–0.6	1.3–2.0	0.3–0.6
T-tail	—	—	0.7–1.2	0.6–1.0
Compare: wing, transport/ASW	*7–9*	*0.3–0.5*		

Hint: think about what each surface is designed to do, when it must not stall, the Mach number at which it meets compressibility, which way it has to produce lift, and its structure.